# Shayan Muhammad

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## Summary

I am a full-stack builder: I take an idea from the people who will use it to a working product. I gather the requirements, design the database, build the React front end and the Node.js or FastAPI back end, and add LLMs where they actually help. I build RAG pipelines, agentic workflows and computer vision systems in Python and TypeScript, and I contribute to open source.

## Experience

**Full-Stack Builder - Strategy Fellow, Strategy Department, Taleemabad (powered by Orenda) – Islamabad** Jul 2026 – Oct 2026

- Contributed to the development of **Skyline**, the internal OKR and project-management tool used by **150+ employees across 10+ teams**.
- Contributing to feature planning, requirements gathering, and implementation.
- Developing Kanban board and reporting features to improve task management and progress tracking.
- Analysing workflows and translating organisational needs into product requirements.
- Supporting user testing, incorporating feedback, and preparing documentation to facilitate adoption.
- Coordinating with team members to track progress and deliver project milestones.

**AI Engineering Intern,** Codecelix – Remote Jun 2026 – Sep 2026

- Developed AutoShorts AI, an end-to-end pipeline that generates short-form video clips from long-form content using NLP, computer vision and LLM-generated captions.
- Built an AI customer-support assistant that combines sentiment analysis with zero-shot classification (BART-large-MNLI) to triage incoming messages into seven ticket categories and draft automated responses.

## Education

**University of Engineering and Technology (UET) Mardan,** B.S. in Computer Science Expected Jun 2028

- CGPA: 3.69 / 4.0. Dean's List, Academic Year 2024–2025.
- Coursework: Data Structures and Algorithms, Machine Learning, Database Systems, Computer Networks, OOP.
- Prime Minister's Laptop Scheme Awardee (National Merit Selection), 2025. Top 10 Rank, Matriculation Board, 2020.
- Kaggle competitions: Digit Recognizer (MNIST, CNN) and Titanic (Random Forest and XGBoost ensemble).

## Projects

**FraudSense: AI Fraud and Risk Detection Platform** Sep 2026

- Owned the authentication and transaction-ingestion service for a team-built fraud detection platform, implementing RS256 JWT with refresh tokens, role-based access control, Fernet field-level encryption, PCI-DSS card masking and security audit logging.
- Built asynchronous bulk CSV ingestion on Celery and Redis in 1,000-record batches with job-status tracking, and containerised the full stack with Docker Compose.

*Technologies: Python, FastAPI, PostgreSQL, SQLAlchemy, Redis, Celery, Docker, Pytest*

**Factory Compliance Monitor** Jun 2026

- Built a safety compliance system in which an LLM converts a health and safety policy into schema-validated rules and YOLOv8 analyses factory video frame by frame, streaming high and critical violations as real-time alerts with structured compliance reports.

*Technologies: Python, YOLOv8, LLMs, Pydantic, Server-Sent Events, JavaScript*

**Multimodal RAG Pipeline** Jun 2026

- Embedded text and figures from PDFs with CLIP (ViT-B/32) into a shared 512-dimensional space with sub-millisecond FAISS retrieval, and grounded Gemini 2.5 Flash answers in retrieved text and charts to reduce hallucination.

*Technologies: Python, CLIP, FAISS, Gemini 2.5 Flash, LangChain, PyMuPDF*

**DOCUMIND: Local RAG for Personal Documents** Jun 2026

- Built a privacy-first document question-answering system that runs entirely on-device, using two-stage retrieval (ChromaDB similarity search followed by cross-encoder reranking) with Llama 3.2 served through Ollama.

*Technologies: Python, ChromaDB, Ollama, nomic-embed-text, CrossEncoder, Gradio*

**DOCUS AI: Blood Report Analyzer** 2025

- Developed a medical report assistant that extracts lab results from PDFs and analyses them through a multi-model LLM pipeline with provider fallback and retrieval-grounded follow-up chat.

*Technologies: Python, Groq LLMs, RAG, Streamlit, Supabase, pdfplumber*

**Open Source: refined-github (32k+ GitHub stars, PR #10013 merged)** 2026

- Contributed a merged enhancement to **refined-github, a 32k+ star TypeScript browser extension**, extending the hide-inactive-deployments feature so that failed deployments are also hidden from pull request timelines.

## Skills

**Languages:** Python, TypeScript, JavaScript, SQL, C++, C, HTML, CSS, Bash
**Full-Stack:** React, Tailwind CSS, Node.js, Express, FastAPI, Flask, Prisma, SQLAlchemy, REST APIs, JWT, OAuth 2.0, RBAC, Redis
**Databases:** PostgreSQL, Supabase, pgvector, FAISS, ChromaDB, Schema Design, Data Migrations
**AI and LLMs:** OpenAI API, Gemini, Groq, Ollama, LangChain, Hugging Face, CLIP, RAG, Agentic Workflows, MCP, QLoRA
**Machine Learning:** PyTorch, TensorFlow, scikit-learn, XGBoost, YOLOv8, OpenCV, CNNs, Transfer Learning
**DevOps and Tools:** Docker, Docker Compose, Git, GitHub, Pytest, AWS S3, Microsoft Azure, Jira, Notion
**Certifications:** Generative AI: Foundation Models (IBM), Python (U. of Michigan), Azure AZ-900, AWS AI Fundamentals